

Multiple Choice (10 marks)

1. What controls the process of molting in crustacea?
 - (a) Water temperature alone without hormonal influence
 - (b) The age of the crustacean
 - (c) Environmental factors
 - (d) Social interactions with other crustaceans
 - (e) Diet of the crustacean
2. A dominant trait has a fitness of 0.6. The frequency in a population of this trait is 1 in 8,000. What is the mutation rate?
 - (a) 2.5×10^{-5}
 - (a) 1.5×10^{-5}
 - (a) 3.0×10^{-5}
 - (a) 4.0×10^{-5}
 - (a) 5.0×10^{-5}
3. Why is it incorrect to say that man evolved from monkeys? What did he possibly evolve from?
 - (a) Man evolved directly from monkeys
 - (b) Man evolved from a branch of primates that includes the modern lemur
 - (c) Man evolved from Parapithecus
 - (d) Man evolved from a direct line of increasingly intelligent monkeys
 - (e) Man evolved from Neanderthals who were a separate species from monkeys
4. All of the following are examples of connective tissue EXCEPT
 - (a) tendons
 - (b) skin
 - (c) muscle
 - (d) blood
 - (e) cartilage
5. Mutation of homeotic cluster genes often results in which of the following developmental defects in *Drosophila*?

- (a) Complete absence of segmentation
- (b) Hindrance in the development of the nervous system
- (c) Inability to undergo metamorphosis
- (d) Absence of a group of contiguous segments
- (e) Tumor formation in imaginal discs

True / False (10 marks)

Answer True or False

- 6. True or False: Do general practice selection scores predict success at MRCGP?
- 7. True or False: Discharging patients earlier from Winnipeg hospitals: does it adversely affect quality of care?
- 8. True or False: Processing fluency effects: can the content and presentation of participant information sheets influence recruitment and participation for an antenatal intervention?
- 9. True or False: Blunt trauma in intoxicated patients: is computed tomography of the abdomen always necessary?
- 10. True or False: Resection of colorectal liver metastases after second-line chemotherapy: is it worthwhile?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Analyze the courtship behavior of the common tern, specifically the male's presentation of fish, and discuss the broader significance of courtship rituals in reproductive success among avian species.
- 12. Discuss the mechanisms by which the biological clock of fiddler crabs can be reset to induce a reversal in their color change pattern, considering environmental and physiological factors.

End of Paper